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Automatic Attendance Detection: Face Recognition

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SUMMARY

This intelligent system project produced an Automatic Attendance Detection system that checks attendance using facial detection. The machine learning implemented in this project are Python, and OpenCV in the development of this system and ran in Pycharm. OpenCV assisted us with analysing students' faces. For instance, image detection, feature extraction, and object detection. Python will be used to program this project.

This system aims to help educators in taking the attendance more efficiently and precisely. This system will detect and identifies the geometry and structure of the students' faces based on the data. Hence, it will be hard for the students to cheat or sign for others which happens a lot when using the traditional method of signing for attendance.

1 INTRODUCTION

1.1 Background

Marking attendance in the paper sheets did not seem applicable for these days. Manual attendance signing has been proven to be difficult and time-consuming and the students are more likely to cheat. Cheating for attendance is very common among students especially during online classes, where teachers are hardly to monitor every single student. Students are no longer physically present in the classroom. Attendance also becomes something important for students to be able to pass nowadays. Therefore, there is a need in approaching stricter method to use in practice. Subsequently, technology can be put into use such as automated or semi-automated attendance marking. Thus, the problem statements of this project are now clear as mentioned previously.

Automated attendance detection is a process that could be implemented by marking through face recognition. Moreover, facial recognition can be an outlook in acquiring students' face to ensure their presence in the class to mark their attendance during online class. Captured images of students in the database will assist the detection easier once their webcams are functioning and able to detect their faces. Once the webcam has captured the student's image, it will be compared with the images in databases for taking attendance.

1.2 Project Objectives

In considering facial recognition into automated attendance detection, the following statements are the objectives of this project:

- 1) To implement machine learning with machine-based visual task in recording the attendance of the class participants by using facial detection.
- 2) To get an accurate number of students expeditiously.
- 3) To promote the usage of technology in education such as attendance taking process.

1.3 Model Objectives

This project will produce a system of automatic attendance detection by fulfilling the objective creating the module. The objectives are:

- 1) To be able to detect unique features of the face from the captured image.
- 2) To compare the features of the captured face to all the images in the database.
- 3) To get the accurate list of students attending the class.

1.4 Expected Output

In this project, the system is planned with the goals and objectives will be achieved. This system will have a database of picture that will be use for detection of the same features from the captured image. When the system is being used for attendance taking, user has to open the camera for capturing the face. The system will detect the face of the user and save it as image. The captured image will be displayed to user. The captured image will be checked so that it will meet the requirement of a good picture. The image will be processed, extract the information of measurement on the face and select the features based on the coding. Then, the features will be compared to the images in the database.

The system will choose the closest features measurement of the captured image when comparing with the image in the database. The chosen image will give the person's name that have same features. The information related to the person will be recorded as attendance and store it in the list of attendance for the day. The date and time in real time will be recorded during the capturing image for attendance. This is done to ensure that the attendance taking is done during class time only. At the end of the program, the expected output that will be seen is the list of attendance will be displayed to show the proof of attendance has been recorded.

2 LITERATURE REVIEW

2.1 Introduction to Automatic Attendance Detection

Automatic Attendance Detection is a web-based system for recording attendance for entire organization. This approach also saves computational power since we do not have to detect the face after transforming to a new frame in the real-time video sequence (Bhattacharya et al., 2018). On the other hand, traditional attendance recording requires manually recording, calling names, and signing on the paper, storing the data, and monitoring students' attendance history. The automatic attendance system detects students' presence and sign them up into their attendance log during a real-time online meeting for a better accuracy. Thus, it is important that the process for taking attendance is accurate, convenient, and quick so that it does not take up a lot of time.

2.2 Causes and Implication of Automatic Attendance Detection

To come up with this machine learning project, the causes and implication of Automatic Attendance Detection must be concerned. It is important so that the project can be done successfully, and the system will work as planned.

Bettadapura (2012), in his literature review about the impact of face detection, noted that:

Automatic face expression recognition systems find applications in several interesting areas. With the recent advances in robotics, especially humanoid robots, the urgency in the requirement of a robust expression recognition system is evident. As robots begin to interact more and more with humans and start becoming a part of our live spaces and work spaces, they need to become more intelligent in terms of understanding the human's moods and emotions. Expression recognition systems will help in creating this intelligent visual interface between the man and the machine (p.2)

2.2.1 Causes of Automatic Attendance Detection

One of the issues that led to the discussion is “the accuracy of the data collected is the biggest issue. This is because the attendance might not be recorded personally by the original person, in another words, the attendance of a particular person can be taken by a third party without the realization of the institution which violates the accuracy of the data” (Suresh et al., 2020). Undoubtedly, huge data can be a burden and may lead to dishonesty from different parties. As it happens, a machine learning technique would help.

Next, in accordance of the causes of this project, time constraint and inefficacious of manual attendance system must be discussed.

Varadharajan et al., 2016 described how manual attendance is time-consuming and ineffective, and how this manual attendance problem must step up:

But these methods consumes a lot of time. Mostly student attendance is taken with the help of attendance sheet given to the faculty members. This consumes a lot of work and time. We do not know whether the authenticated student is responding or not. Calculation of consolidated attendance is another major task which may cause manual errors. In some other cases the attendance sheet may become lost or stolen by some of the students. To overcome such troubles, we are in need of automated attendance management system. (p.1)

In addition, the contribution of the causes of automatic attendance detection, the correlation technology utilization and online learning is required to be commercialized.

The researchers studied how the flexibility should be emphasized into online learning in utilising the technology for online classes:

This growing dependency on technology has driven educators to reconsider their traditional teaching methods and invest in the latest technologies and tools. There are various applications, devices and uses to teach and learn from ICT,

such as e-learning, educational software and the Internet. In terms of distance-based education, e-learning or learning technology remains a widely used term (Qazi, et al., 2021, p.45539).

In general, due to the accuracy of the data for attendance that had been collected manually, time and energy constraint, and the utilisation of technology in elearning, machine learning has to be involved in order to overcome the causes of this project.

2.2.2 Implication of Automatic Attendance Detection

When the involvement of machine learning is usually raised to be impactful, there must be some implications of this machine learning project. Bora et al., (2014) general structure and ideas of recognition, important issues and factors of human faces. In this work a discussion on some critical techniques and algorithms along with a comparison are presented (p. 526).

First, face recognition is a good way to be implemented to the system to automatically taking attendance efficiently by the machine learning interpretation. However, misinterpretation of data may occur. During the process of face detection, the captured image will be compared to the image in the database. Clearly, some people may have similar or distinctive features. The extraction of faces for face detection may the attendance-taking process expeditiously. In reality, these assumptions are often hard to be verified. Therefore, a practical solution for selecting an appropriate model for a given classification task is to experimentally compare these algorithms (García Amaro et al., 2012, p. 214). More detailed algorithm needed to be implemented to reduce the problem of similar faces. Although many of the same techniques (e.g. linear models) can be applied to both predictive and causal queries, they require distinct processes for their application and subsequent interpretation of results (Arnold, et al., 2020). According to (Howse, 2013) “some means of abstracting image detail is useful in producing stable classification and tracking results” (p. 62). Regtech (2021) noted that “face recognition is a technology capable of identifying or verifying a subject through an

image, video or any audiovisual element of his face. Generally, this identification is used to access an application, system or service. Hence, algorithms can effectively solve the data-related problems and as long as we believe in the computational intelligence, and doing it right at the same time, it may work.

On the other hand, face recognition in automated attendance detection includes huge storage requirements and potential privacy issues. Large data is not common in machine learning, in ensuring the accuracy of the data. However, the requirements must be fulfilled, especially in performing face recognition for attendance recognition. Chaves, et al., (2020) noted that “computationally demanding and may not be suitable for applications that analyse large amounts of data and require real-time performance” (p.3). Notably, the collection of data is to be known as lengthy, but the result did pay off. Nevertheless, the security measure can be concerning. Fox (2020) described three major privacy concerns that accompany facial recognition are the misuse of biometrics and facial recognition, security breaches, and improper data storage and sharing (para 2). Thus, data has to be protected to avoid inconvenience.

Above all, the implication of automatic attendance detection could be the misinterpretation of data, and huge storage requirements and potential privacy issues.

2.3 Machine Learning Used

Face recognition by using images was required to apply face detection. Subsequently, OpenCV is one of the libraries that is required for image recognition which is the main objective for this project.

2.3.1 OpenCV

OpenCV is an open-source computer vision software library that is used in machine learning. OpenCV is used for all sorts of image and video analysis, like facial recognition and detection, license plate reading, photo editing, advanced robotic vision, optical character

recognition, and a whole lot more (Sentdex, 2015). The library contains more than 2500 advanced algorithms including an extensive set of both typical and state-of-the-art computer vision and machine learning algorithms (Bussa et al., 2020). Real-time face recognition can be applied using OpenCV and the data can be collected for attendance.

3 EXPERIMENTAL SETUP

3.1 Dataset Description

One of the really precious resources is data. As our project is related with AI face recognition, the majority of the data we used for our project was images. These images are the images of persons. For a better outcome and identification when we run the code, the images are in JPG file and preferably in high-definition format. As images are non-numerical data, they are classified as qualitative data. Furthermore, since there is no order in it, it also falls under the nominal data category.

We collect the data by using the search engines, Google. We browse for celebrity photos and choose up to four of them to serve as our dataset in the code. We only take photographs that are permissible for educational purposes and do not require attribution. When the programme is executed, the photographs will be kept in the programme so that it may read the data and analyse the facial features of these persons in order to recognise them. Facial features include eyes, eyebrows, mouth, nose, and jaw as the basic (Diego-Mas et al., 2020).

3.2 Data Pre-processing

3.2.1 Data Cleaning

While searching for and gathering photographs to include in our dataset, we came across several that were fuzzy, had gloomy lighting, or were broken. As a result, we will eliminate the photographs that do not meet our criteria for a good picture and replace them with new ones. This is due to the fact that a lousy shot will reduce the program's accuracy and may

disrupt the efficiency of the program's purpose. An excellent photograph must clearly show all of the features of the face.

3.2.2 Data Transformation

Arthur in his article said that “Most colour photographs are made up of three interconnected arrays” on coloured image structure (Arthur, 2019). Those interconnected arrays are known as RGB values where R represents Red, G represents Green and B represents Blue. The RGB colours were referred to as primary colours (Pedamkar, 2020). RGB colour model's primary usage is for image perception, interpretation, and display in electronics devices. The pictures will be saved in a database after they have been processed. As a result, in our project, the dataset will be transformed to RGB format using OpenCV after they are imported to facilitate comparison while the code is executing. Attached below is how it sees during the transformation process.

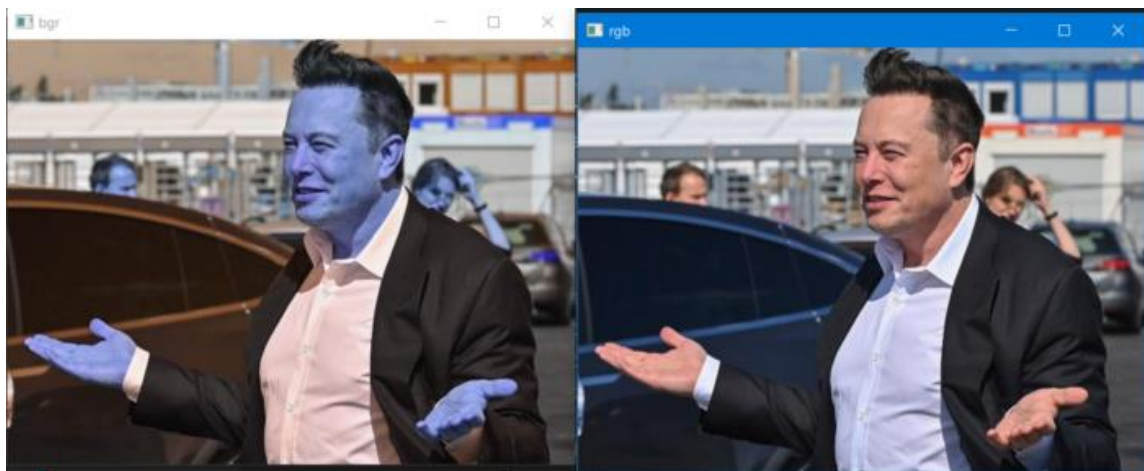


Figure 1: Image transformation example.

4 METHODOLOGY

4.1 System Development

When developing the system, we must first understand its flow to ensure that we are properly planning and building it. For this project we use PyCharm IDE as the medium to code the system, Python as the programming language and OpenCV for the library. As said, we used the OpenCV library for our project as it is the most fitting for our objectives. Cmake, dlib, face_recognition, numpy and opencv-python are the required packages installed in the PyCharm.

Before we begin with the coding, we must create a database containing the images of the students we wish to recognise later. However, to simplify the process of data seeking for our project's purposes, we just employ photographs of famous celebrities because they are easy to obtain and widely available. These photographs will then be used as our dataset in our coding. Only 1 picture per person is already enough for the system to do image recognition.

For the next step in the code, the webcam will capture the real time image of the person, crop and resize it and then also convert the images to RGB values and encode it in the system. This is the same step that is done to the images of the dataset that have been imported earlier. These encoded values then will be compared with each other. If it matches, it means that the system recognized the person, and the name of the said person will appear on the screen. The name and time will also be in the excel file for the lecturer's perusal.

As for the people who are not in the database or dataset. The system will display an "Unknown" text at the person's face. This shows that the system does not recognise the person as there is no data of him or her in the system to be compared with. A flowchart, the pseudocode and the code of the system are included in this report for better understanding.

4.2 Flowchart

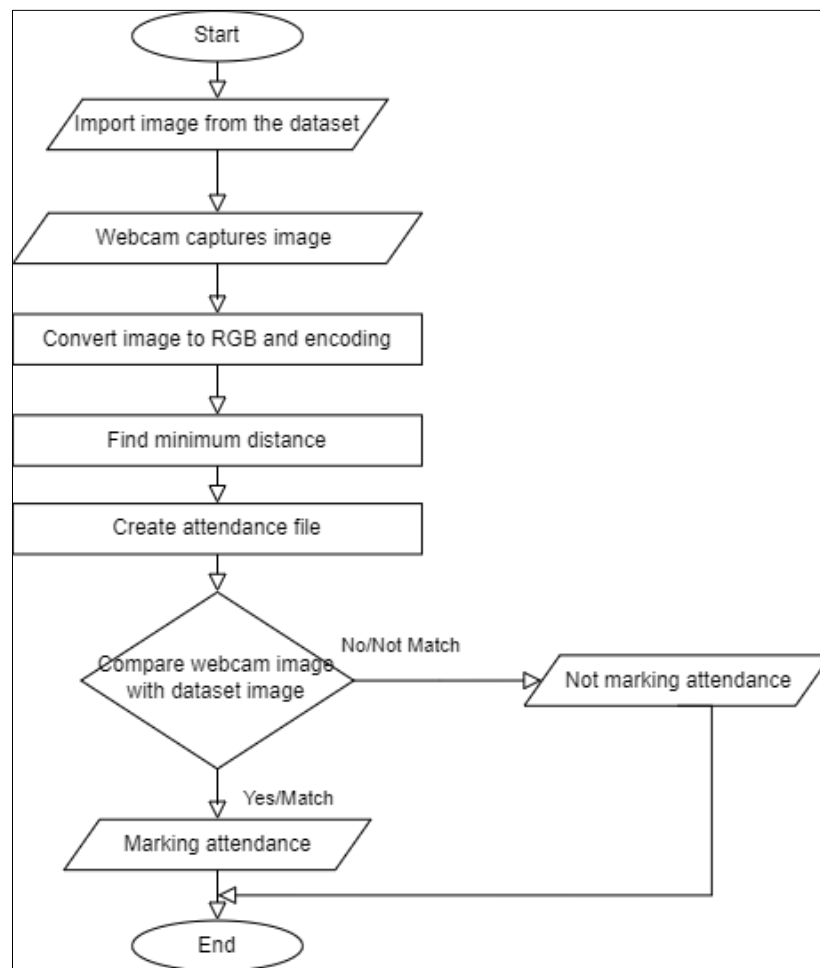


Figure 2: Flowchart of Face Recognition.

4.3 Pseudocode

1. Import image from dataset
2. **While** Loop Capture the image from the webcam
3. Convert image into RGB
4. Encoding image using the face_encodings() function.
5. Compute and find minimum distance.
6. Create attendance file in CSV format.
7. **If** matches **Then**
 - Print name
 - Mark attendance, name, date, time**Else**
 - Print unknown
8. **End**

4.4 Code

```
# Importing
import face_recognition
import cv2
import numpy as np
import os
from datetime import datetime

#Importing Images from the directory
path = 'ImagesAttendance'
images = []
classNames = []
myList = os.listdir(path)
print(myList)
# Split Name
for cl in myList:
    curImg = cv2.imread(f'{path}/{cl}')
    images.append(curImg)
    classNames.append(os.path.splitext(cl)[0])
print(classNames)

#Compute Encodings from the dataset
def findEncodings(images):
    encodeList = []
    for img in images:
        img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
        encode = face_recognition.face_encodings(img)[0]
        encodeList.append(encode)
    return encodeList

#Marking Attendance
def markAttendance(name):
    with open('Attendance.csv', 'r+') as f:
        myDataList = f.readlines()
        nameList = []
        for line in myDataList:
            entry = line.split(',')
            nameList.append(entry[0])
        if name not in nameList:
            now = datetime.now()
            dtString = now.strftime('%H:%M:%S')
            f.writelines(f'\n{name},{dtString}')

encodeListKnown = findEncodings(images)
print('Encoding Complete')

#Matches images with webcam
cap = cv2.VideoCapture(0)

while True:
    success, img = cap.read()
    imgS = cv2.resize(img, (0,0), None, 0.25, 0.25) #resize the webcam images
    imgS = cv2.cvtColor(imgS, cv2.COLOR_BGR2RGB) #convert into RGB

    facesCurFrame = face_recognition.face_locations(imgS) #face location
    encodesCurFrame = face_recognition.face_encodings(imgS, facesCurFrame) #encodings

    #compare faces
```

```

for encodeFace, faceLoc in zip(encodesCurFrame, facesCurFrame):
    matches = face_recognition.compare_faces(encodeListKnown, encodeFace)
    faceDis = face_recognition.face_distance(encodeListKnown, encodeFace)
    #print(faceDis)
    matchIndex = np.argmin(faceDis)

    if matches[matchIndex]:
        name = classNames[matchIndex].upper()
        #print(name)
        # Rectangle at the face
        y1,x2,y2,x1 = faceLoc
        y1, x2, y2, x1 = y1*4,x2*4,y2*4,x1*4
        cv2.rectangle(img,(x1,y1),(x2,y2),(0,255,0),2)
        cv2.rectangle(img,(x1,y2-35),(x2,y2),(0,255,0),cv2.FILLED)
        cv2.putText(img,name,(x1+6,y2-
6),cv2.FONT_HERSHEY_COMPLEX,1,(255,255,255),2)
        markAttendance(name)

# If the face is not match
else:
    name = 'Unknown'

    # print(name)
    # Rectangle at the face
    y1, x2, y2, x1 = faceLoc
    y1, x2, y2, x1 = y1 * 4, x2 * 4, y2 * 4, x1 * 4
    cv2.rectangle(img, (x1, y1), (x2, y2), (0, 255, 0), 2)
    cv2.rectangle(img, (x1, y2 - 35), (x2, y2), (0, 255, 0), cv2.FILLED)
    cv2.putText(img, name, (x1 + 6, y2 - 6), cv2.FONT_HERSHEY_COMPLEX, 1, (255, 255,
255), 2)

    cv2.imshow('Webcam',img)
    cv2.waitKey(1)

```

5 RESULT

The results for our project are as follows. The system will compare the face of the person from the dataset and image that captured by the webcam. If the image from the dataset and captured image from Webcam is match the system will display their names. The attendance will be marked by listing the names and time in attendance file. However, if the image from the dataset and captured image from webcam is not a match the system will display 'Unknown', and the attendance will not be marked in the attendance file.

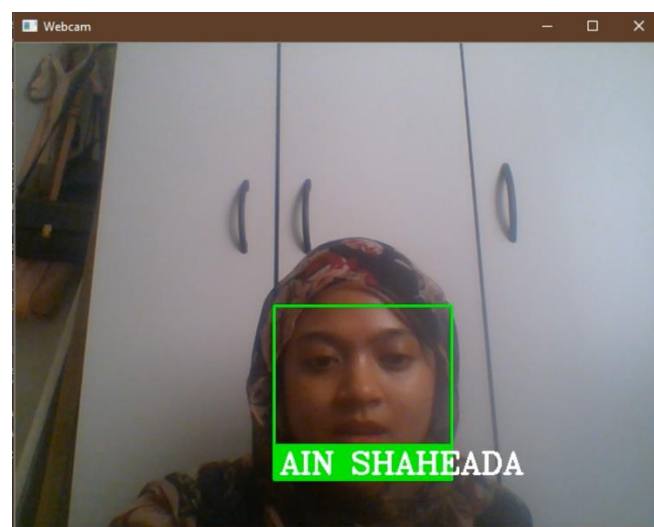


Figure 3: Result of Ain Shaheada's face recognition



Figure 4: Result of Mark Zuckerberg's face recognition

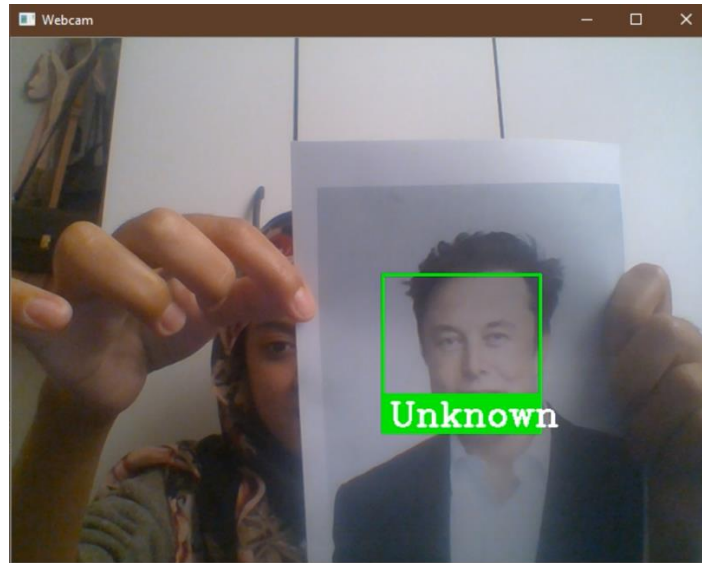


Figure 5: Result of the face that is not in the dataset

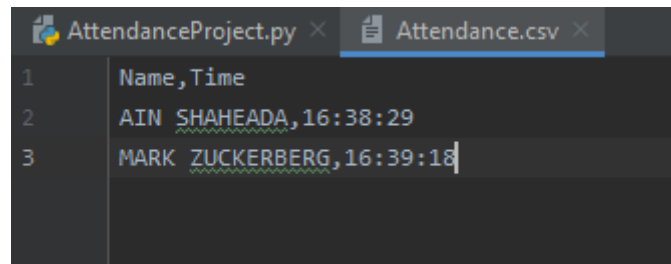


Figure 6: Recorded attendance

6 CONCLUSION

Taking all of this into account, machine learning can be applied in creating systems that will make work easier. Manual attendance taking is a hassle process and easy to cheat. By applying machine learning in creating automatic attendance system, it makes the attendance taking process a lot easier and students cannot cheat in the attendance. The automatic attendance taking is done by using webcam to capture the face of the students. OpenCV was used in order to be able to do face detection and face recognition. The attendance will be taken after the system able to detect all the features on the face and compare it to the image in database.

Using OpenCV is the best decision that we have made as OpenCV is an excellent tool for image analysis and computer vision tasks. The open-source library contains the task that we want to perform in this project which is face-detection. It is also easy to understand on how to use it as there are many exploration have been done on the functions and algorithms in OpenCV. Furthermore, it supports the python programming language making it easier to apply it in our project since we are using python for the development of our project.

Next, the advantage of adopting an open-source library is that it allows students like us to gain access to the library and develop a face recognition attendance marking system like this. This system will be of tremendous use to schools all around the world in terms of attendance tracking. It will certainly help ease the burden of the educators that usually mark attendance manually by calling and ticking the names of their students one by one. Not to forget, it also fastens the time taken for attendance marking as the system will mark attendance faster and more efficient than the traditional way.

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